and correspondingly a unique function. The structure of proteins can be best understood by considering them in four hierarchical levels as described in figure 3.7

3.3.1 The primary structure of proteins

The amino acid sequence of a protein is known as its primary structure. Knowing the primary structure for a protein is important because even small changes (due to mutations) in the primary structure can lead to improper folding and hence impairment or complete loss of function.

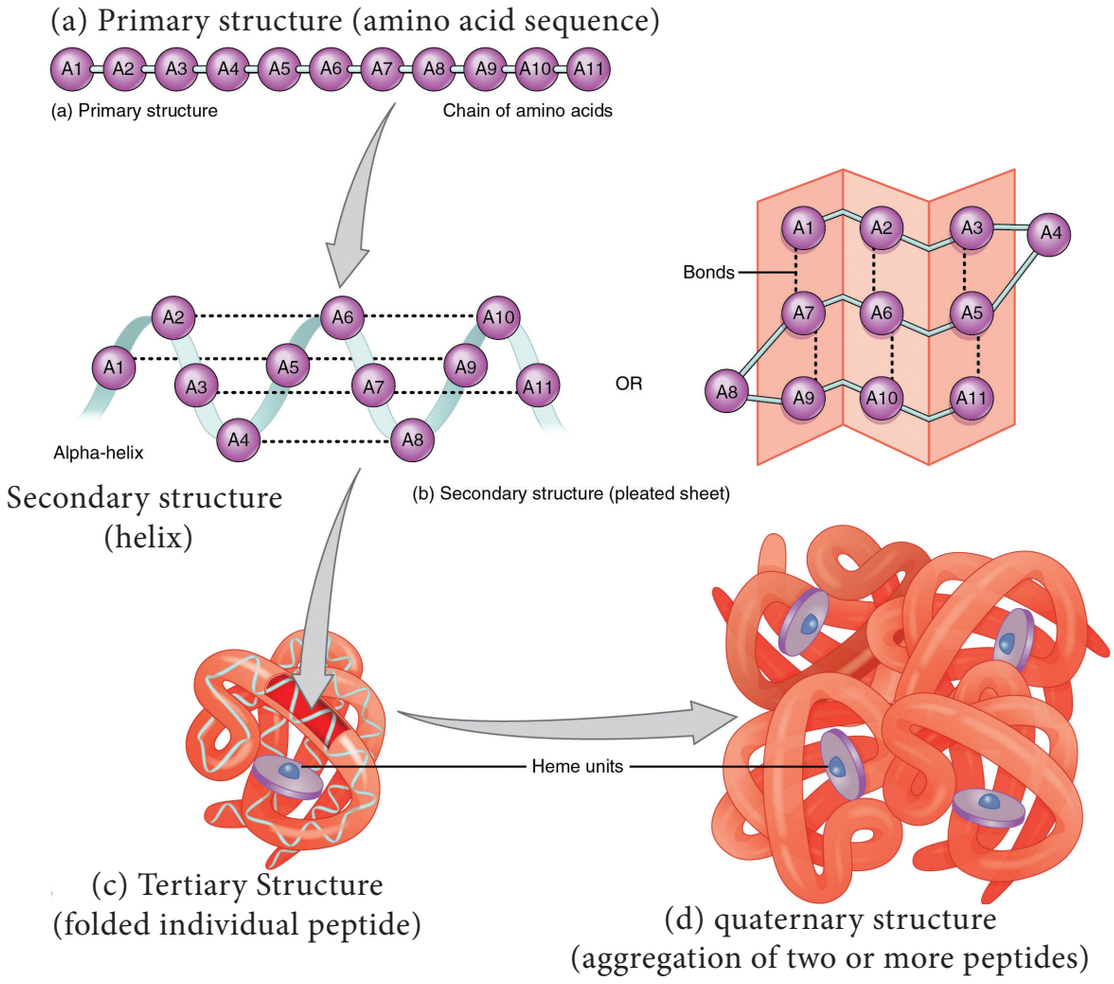


Figure 3.7 Structural organization of proteins

Peptide bonds

The amino acids in protein are covalently linked together to form peptide bonds. Peptide bonds are amide linkages between the α carboxyl group of one amino acid and the α amino group of another amino acid. For example, serine and alanine can form a peptide called serylalanine as described in figure 3.8. Since two amino acids are joined together, this molecule is known as a dipeptide. If many amino acids are joined together in the same way to form a single chain, such a chain is known as a polypeptide. The atoms excluding

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